# Edge-Driven Federated System for UPI Fraud Detection

*B.Tech. Project (CS400) Report*
*submitted in partial fulfilment of the requirements for the award of the degree of*

***Bachelor of Technology***

*by*

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## DECLARATION

I hereby *declare* this Project entitled **Edge-Driven Federated System for UPI Fraud Detection** that is being submitted to **Indian Institute of Information Technology Guwahati, Assam**, in partial fulfilment for the requirements of the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** in the department of **Computer Science and Engineering**, is a *genuine report of the work carried out by me*. The material contained in this report has not been submitted at any other University or Institution for the award of any degree.

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Place: IIIT Guwahati, Assam (Signature of the Student)

Date: December 1, 2025 Computer Science and Engineering

**Abstract**

With the rapid growth of UPI-based digital transactions, fraud detection systems must become more accurate while also preserving user privacy. Traditional centralized models require sensitive data to be collected and stored in a single location, increasing the risk of breaches and violating privacy principles. This project proposes an Edge-Driven Federated Learning system for UPI fraud detection, where data remains at the client side and only model updates are shared for aggregation.

In the centralized setup, multiple machine learning models including MLP, KNN, XGBoost, Logistic Regression, and Random Forest were trained and evaluated on a unified dataset to establish benchmark performance. Among these, the MLP model showed strong potential and was selected as the primary architecture for the implementation of Federated Learning (FL). A Federated MLP and Federated Logistic Regression were then developed, where multiple clients trained local models on their private datasets and sent weight updates to a central aggregator over multiple communication rounds.

To simulate a real-world distributed environment, the federated system was deployed and tested using AWS EC2 instances, with each client running on an edge node and the server performing parameter aggregation. The system was evaluated on metrics such as accuracy, precision, recall, and F1-score. The results demonstrated that although centralized models initially outperformed the federated ones, careful tuning of hyperparameters, learning rate, communication rounds, and federated aggregation strategies significantly improved the federated models, bringing their performance close to that of the centralized models.

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# 1 Introduction

## 1.1 Overview

Millions of users rely on UPI daily for instant, cashless payments due to its accessibility, convenience, and speed. However, this massive adoption has also resulted in a significant rise in fraudulent activities, making fraud detection a critical area of concern. Conventional fraud detection methods rely heavily on centralized machine learning systems, where all transaction data is collected and stored in a single repository for analysis and training. Although these methods often produce accurate results due to the availability of aggregated data, they raise serious concerns regarding user privacy, data security, and regulatory compliance.

To address these limitations, this project proposes an Edge-Driven Federated Learning system for UPI fraud detection. In this system, the data remains on the client's device, and only model updates are shared with the central server for aggregation. The main model used in the federated setup is a Multi-Layer Perceptron (MLP), supported by Logistic Regression for comparison. The results from the federated system are then compared with several centralized machine learning models such as KNN, XGBoost, Logistic Regression, Random Forest, and MLP.

## 1.2 Challenges

UPI fraud detection presents several technical and practical challenges:

- Data privacy concerns: Financial transaction data is highly sensitive, and centralizing it poses risks such as data leakage, misuse, and violation of privacy regulations.
- Imbalanced datasets: Fraudulent transactions typically form only a small portion of the total dataset, making it difficult for models to learn patterns effectively.
- Imbalanced datasets: Fraudulent transactions typically form only a small portion of the total dataset, making it difficult for models to learn patterns effectively.
- Heterogeneous data distribution: In a federated setup, data across different clients may vary significantly, affecting model convergence and performance.
- Communication overhead: Federated Learning requires multiple rounds of communication between clients and the server, which can increase latency and computational load.
- Performance gap: Centralized models often outperform federated models due to access to complete datasets, creating a challenge in bridging the performance gap while maintaining privacy.

## 1.3 Motivation

The main motivation behind this project is the growing need for secure and privacy-preserving fraud detection systems in digital financial platforms like UPI. As cybercrimes continue to increase, relying solely on centralized systems is no longer sufficient due to their vulnerability to breaches and privacy violations.

Federated Learning offers a promising alternative by ensuring that sensitive data never leaves the user's device. This project is driven by the idea that privacy and performance do not have to be mutually exclusive. By carefully tuning the federated MLP model, optimizing training rounds, and improving aggregation techniques, it is possible to achieve performance that is comparable to centralized models while maintaining data privacy.

Additionally, deploying the system on AWS EC2 helped in understanding the real-world feasibility of scalable federated architectures. This makes the solution both technically relevant and practically applicable in real financial systems.

### 1.4 Problem Statement

Traditional centralized fraud detection systems require transaction data from multiple users to be stored and processed in a central server, exposing sensitive financial information to potential security threats and privacy violations. Given the rising number of UPI fraud incidents, there is a critical need for a secure, efficient, and privacy-preserving fraud detection system.

Therefore, the problem addressed in this project is: "To design and implement an Edge-Driven Federated Learning system for UPI fraud detection that preserves data privacy while achieving classification performance comparable to centralized machine learning models."

## 2 Related Works

One of the studies introduced a federated learning (FL) approach for detecting fraudulent UPI transactions by enabling decentralized model training across participating entities. In this approach, only model updates are shared for aggregation, while the original transaction data remains on the local systems, reducing direct data exposure [1]. However, the work remains largely conceptual and experimental in a controlled environment. Edge-based deployment and testing using real distributed infrastructure such as AWS EC2 instances were not performed, leaving the practical scalability and real-world feasibility of the framework unverified.

In another study, a machine learning-driven fraud detection system for UPI transactions was developed using the XGBoost algorithm as the core classifier. The authors demonstrated that XGBoost is well-suited for handling complex and imbalanced financial datasets and achieved an accuracy of 98.2. To handle the inherent class imbalance in fraud data, the study employed the Synthetic Minority Over-sampling Technique (SMOTE), which helped generate synthetic samples of the minority class and improve the model's ability to recognize fraudulent patterns [4]. Additionally, Principal Component Analysis (PCA) was applied for dimensionality reduction, improving feature representation and reducing redundancy in high-dimensional datasets.

The importance of data balancing techniques is also highlighted through references to Random Over Sampling (ROS), Random Under Sampling (RUS), and advanced oversampling methods like DeepSMOTE, which combines deep learning with SMOTE to generate high-quality synthetic data for the minority class [4]. Experimental evidence shows that such sampling techniques significantly improve recall, precision, and F1-score when dealing with extremely imbalanced fraud datasets, which is a dominant characteristic of financial transaction data [4].

Further improvements in fraud detection performance were explored through ensemble learning and neural networks. One study implemented an ensemble-based framework combined with neural networks to increase detection accuracy and enhance generalization across diverse transaction patterns [2].

While the majority of UPI fraud detection studies focus on centralized architectures, the application of Federated Learning in other sensitive domains, such as medical image segmentation, has demonstrated its effectiveness in privacy-preserving collaborative model training. The cloud-based federated learning framework for MRI segmentation confirms that decentralized learning approaches can achieve competitive performance while maintaining data locality and confidentiality [3]. This validates the suitability of federated learning for similarly sensitive financial transaction data, such as UPI records.

# 3 System Architecture

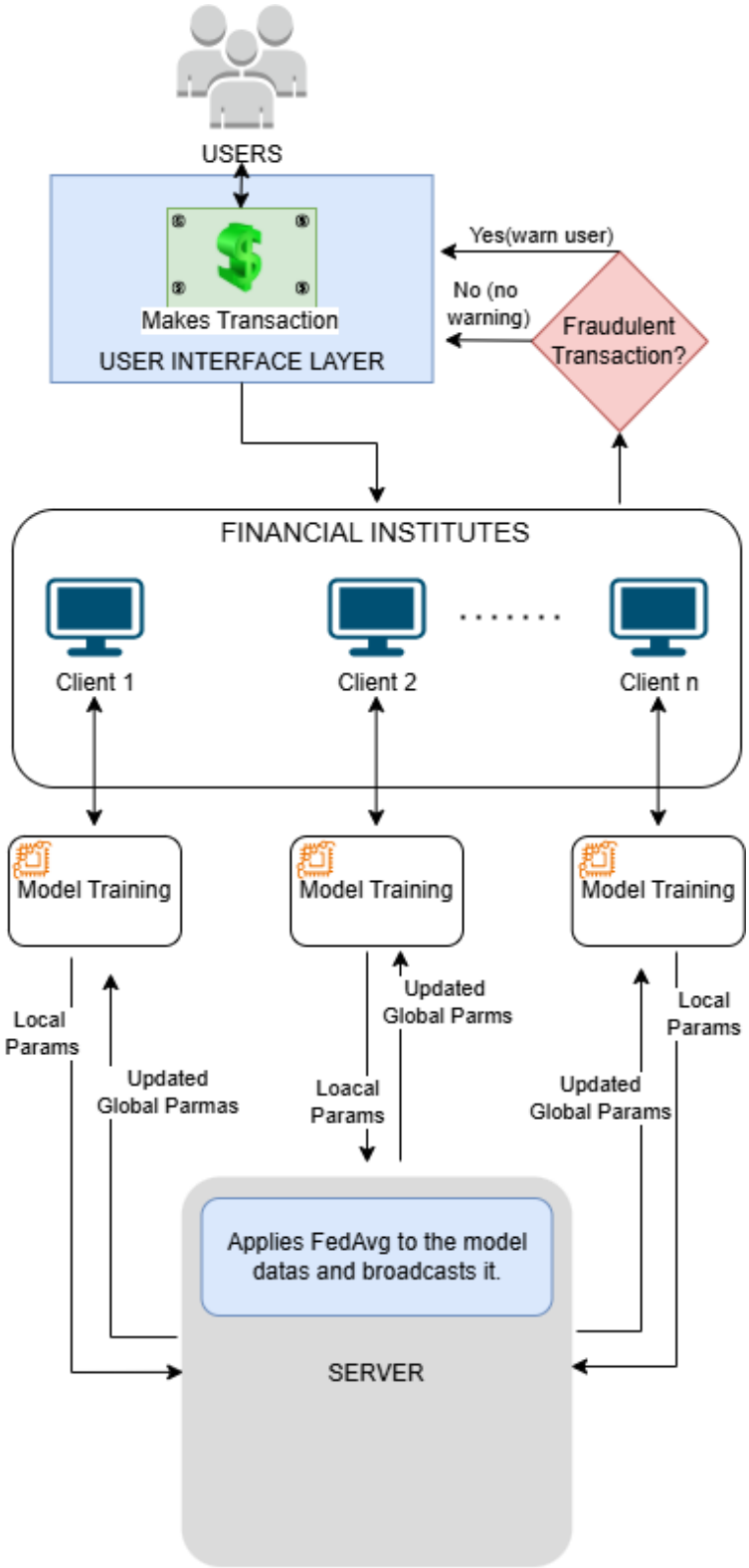


Figure 1: System Architecture

# 4 Algorithms

## 4.1 Centralized Learning Models

In the centralized setting, all transaction data is combined into a single dataset and used for training and evaluation. The following supervised learning algorithms were implemented and compared:

- Multi-Layer Perceptron (MLP)
- Logistic Regression (LR)
- K-Nearest Neighbors (KNN)
- Random Forest (RF)
- XGBoost (XGB)

Among these, the MLP model was selected as the primary learning model for the federated implementation, due to its ability to capture non-linear relationships in transactional data.

## 4.2 Centralized MLP Architecture

The MLP architecture used in this work consists of five layers and is defined as follows:

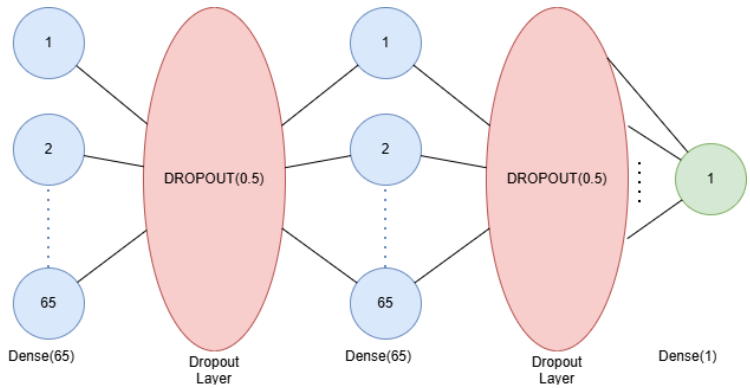


Figure 2: MLP

All features are normalized using StandardScaler before training. The model is optimized using the Adam optimizer and binary cross-entropy loss.

**Algorithm 1** Centralized MLP Training

**Input:** Dataset $D = \{(x_i, y_i)\}$, epochs 200, batch size 32, learning rate 0.001
**Output:** Trained MLP model $M$
Normalize features of $D$ using StandardScaler
Split $D$ into training and testing sets
Initialize MLP model
**for** epoch $= 1$ to $E$ **do**
  **for** each mini-batch $(x_b, y_b)$ in training set **do**
    Forward propagate $x_b$ through the network
    Compute loss using binary cross-entropy
    Backpropagate the error
    Update weights using Adam optimizer
  **end for**
**end for**
Evaluate the model on test set using Accuracy, Precision, Recall and F1-score

## 4.3 Federated Learning Model

In the federated setting, the training data remains distributed among multiple clients. Each client trains the model locally and only the model parameters are shared with a central server for aggregation.

In this work, two clients are considered:

$$Client_1 \to D_1, \quad Client_2 \to D_2$$

The server follows the Federated Averaging (FedAvg) algorithm for global model aggregation.

## 4.4 Server-Side Algorithm

**Algorithm 2** Server-Side Federated Learning Process

**Input:** Number of rounds $R$, number of clients $K = 2$
**Output:** Final global model $W^R$
Initialize global model weights $W^0$
**for** round $r = 1$ to $R$ **do**
  **for** each client $k \in \{1, 2\}$ **do**
    Send global weights $W^r$ to client $k$
  **end for**
  **for** each client $k \in \{1, 2\}$ **do**
    Receive locally trained weights $W_k^r$ from client $k$
  **end for**
  Aggregate using Federated Averaging:

$$W^{r+1} = \frac{1}{K} \sum_{k=1}^{K} W_k^r$$

**end for**
Return final global model $W^R$

## 4.5 Client-Side Algorithm

Each client trains the model on its private dataset and sends the updated parameters to the server.

**Algorithm 3** Client-Side Local Training

**Input:** Local dataset $D_k$, global weights $W^r$, local epochs $E$
**Output:** Updated weights $W_k^r$
Receive global weights $W^r$ from server
Apply StandardScaler to dataset $D_k$
**for** epoch $= 1$ to $E$ **do**
    Perform forward propagation on $D_k$
    Compute binary cross-entropy loss
    Perform backpropagation
    Update weights using Adam optimizer
**end for**
Return updated weights $W_k^r$ to server

# 5 Experimentation and Results

This section presents the experimental setup, evaluation metrics, and the results obtained from both centralized and federated learning models. The experiments were conducted with a particular focus on optimizing the Federated MLP model and comparing its performance with centralized models.

## 5.1 Experimental Environment

The experiments were carried out in two different environments:

- **Centralized environment:** All data was combined and trained on a single machine.
- **Federated environment:** A server–client architecture was implemented using AWS EC2 instances, where two clients trained the model on local datasets and sent only model parameters to the central server.

AWS EC2 instances were used to simulate edge nodes (clients) and a central aggregation server. Each client had its own local dataset and performed independent training before sending model weights to the server.

## 5.2 Evaluation Metrics

The performance of the models was evaluated using the following standard metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- AUC-ROC Curve

These metrics were chosen because fraud detection is a highly imbalanced classification problem, and accuracy alone is not sufficient for evaluating real performance.

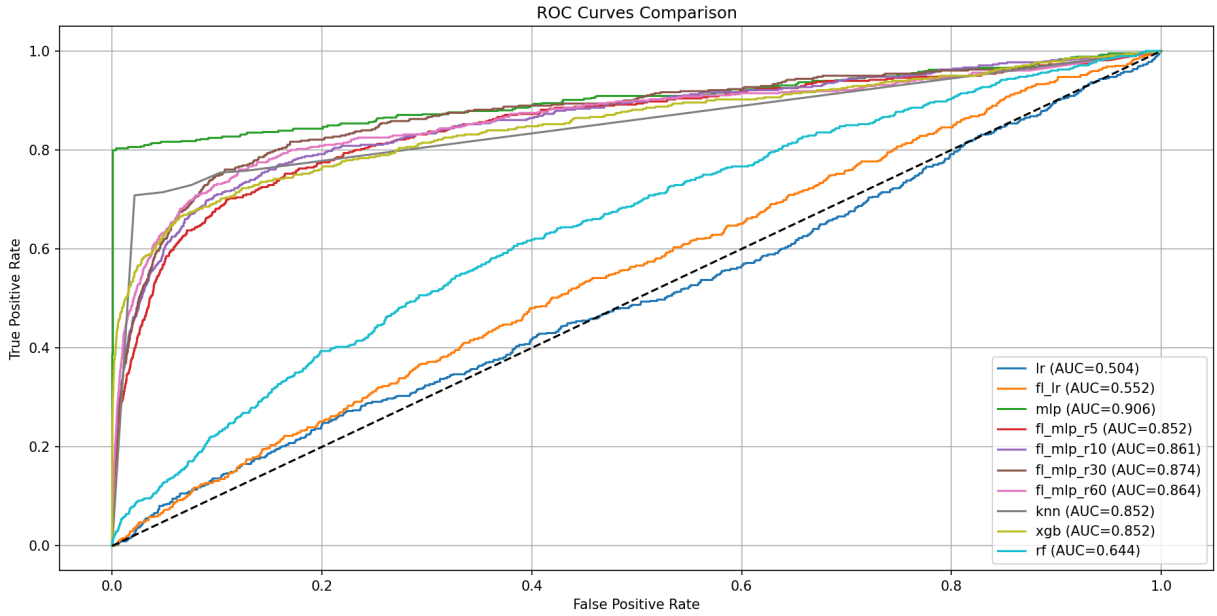


Figure 3: AUC-ROC Curve for Centralized and Federated Models

## 5.3 AUC-ROC Curve Analysis

The Area Under Curve (AUC) of the Receiver Operating Characteristic (ROC) was used to measure the model's ability to distinguish between fraudulent and legitimate transactions.

From the figure, it can be observed that the centralized models achieved a higher AUC score. However, with multiple communication rounds, the Federated MLP model showed a progressive increase in AUC, reducing the performance gap.

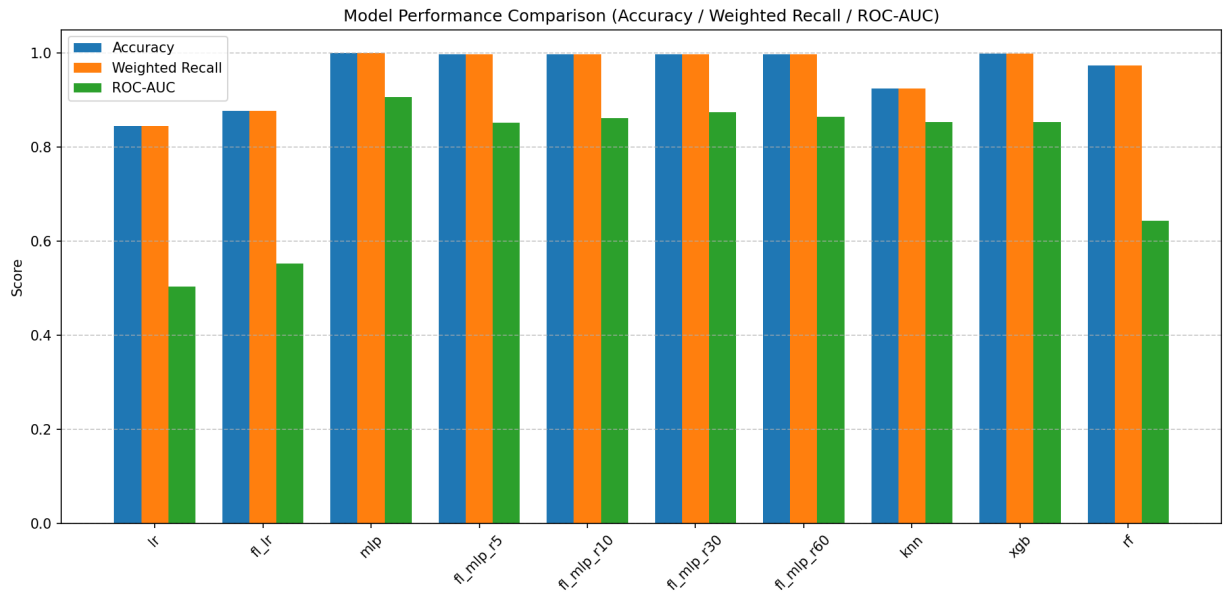


Figure 4: Comparison of Accuracy, Precision and Recall

The centralized models achieved strong overall accuracy, with XGBoost and Random Forest performing particularly well. Among federated models, the Federated MLP outperformed Federated Logistic Regression, indicating its stronger ability to capture non-linear patterns even in a distributed environment.

Although the Federated MLP shows slightly lower performance compared to the centralized MLP, the

Table 1: Performance Comparison of Centralized and Federated Models

| Model | Accuracy | ROC-AUC | Precision | Weighted Recall |
|---|---|---|---|---|
| Logistic Regression (LR) | 0.844124 | 0.503552 | 0.002405 | 0.844124 |
| Federated LR (FL-LR) | 0.876924 | 0.551845 | 0.002562 | 0.876924 |
| MLP (Centralized) | 0.998936 | 0.905563 | 0.693141 | 0.998936 |
| FL-MLP (R = 5) | 0.996484 | 0.851763 | 0.114120 | 0.996484 |
| FL-MLP (R = 10) | 0.996836 | 0.860918 | 0.142529 | 0.996836 |
| FL-MLP (R = 30) | 0.997216 | 0.874392 | 0.156051 | 0.997216 |
| FL-MLP (R = 60) | 0.996948 | 0.863679 | 0.147132 | 0.996948 |
| KNN | 0.924152 | 0.852151 | 0.018246 | 0.924152 |
| XGBoost | 0.998084 | 0.852200 | 1.000000 | 0.998084 |
| Random Forest | 0.972828 | 0.643594 | 0.006874 | 0.972828 |

R: Rounds

difference is acceptable considering the strong privacy advantages provided by the federated learning approach.

### 5.4 AWS EC2 Deployment

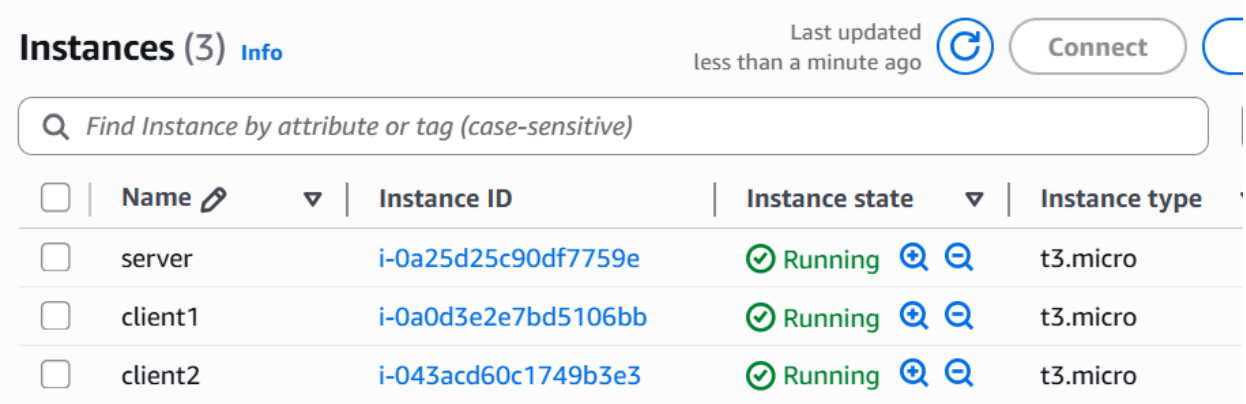


Figure 5: AWS Ec2 instances Running

The server and clients were deployed using separate AWS EC2 instances. The server handled model aggregation while the clients handled local training operations. These instances communicated securely over the network to exchange only model weights during each round of the federated training process.

## 6 Conclusion

This project presented the design and implementation of an **Edge-Driven Federated Learning System for UPI Fraud Detection**, with a primary focus on developing a privacy-preserving alternative to conventional centralized machine learning approaches. The study successfully compared the performance of multiple centralized models, including Logistic Regression, K-Nearest Neighbors, Random Forest, XGBoost, and Multi-Layer Perceptron (MLP), with federated implementations of Logistic Regression and MLP. Among the centralized models, the MLP achieved the best overall performance, with the highest ROC-AUC and precision, demonstrating its strong capability in modelling complex patterns within highly imbalanced financial transaction data. This confirmed its suitability as the main model for the federated environment.

In the federated setting, a two-client architecture was implemented using AWS EC2 instances to simulate edge devices. Each client trained the model on its private dataset and shared only model parameters with

the central server using the Federated Averaging (FedAvg) algorithm. This ensured that sensitive transactional data remained at the source and was never exposed during training. The experimental results showed that the Federated MLP consistently outperformed Federated Logistic Regression across all configurations. As the number of communication rounds increased, the model demonstrated improved convergence, with the best balance between ROC-AUC and precision observed at 30 rounds. Beyond this point, performance gains became marginal, indicating a saturation effect. Although a performance gap still exists between centralized and federated models, this gap was significantly reduced through architectural optimization and increased communication rounds.

Despite achieving slightly lower precision than the centralized MLP, the Federated MLP maintained very high accuracy and weighted recall. This behavior highlights the impact of the extreme class imbalance present in fraud detection datasets and reinforces the importance of using metrics such as ROC-AUC and precision for evaluation rather than relying solely on accuracy. Overall, the results verify that an edge-driven federated approach can effectively be applied to UPI fraud detection. The proposed system achieves a strong balance between performance and data privacy, making it highly suitable for real-world financial applications where data confidentiality is critical. Furthermore, the successful use of AWS EC2 to simulate distributed clients demonstrates the practical feasibility and scalability of the proposed federated framework.

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